

QUEZON CITY SCIENCE GARDEN, Philippines

WMO: 984300

Lat: 14.6452N

Lon: 121.0441E

Elev: 151

StdP: 14.62

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	68.5	70.1	62.4	84.9	76.8	63.8	89.1	76.7	13.3	82.6	11.5	82.7	0.6	0	0.299

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	14.4	95.1	78.7	93.8	78.6	92.6	78.6	81.8	89.9	81.2	89.0	80.7	88.4	4.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.8	155.5	85.8	79.1	152.1	85.1	78.7	149.9	84.6	45.7	90.4	45.1	89.5	44.5	88.8	92.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.0	9.9	8.0	64.2	98.8	3.7	2.1	61.5	100.3	59.4	101.5	57.4	102.7	54.7	104.2		
(5)			61.6	85.3	2.9	2.9	59.6	87.4	57.9	89.1	56.3	90.7	54.2	92.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.9	78.6	79.8	81.8	84.5	85.2	83.7	82.3	82.0	82.1	81.8	81.2	79.7		
	DBStd	2.82	2.18	2.20	2.16	2.17	2.29	2.23	2.17	2.15	2.01	1.78	1.90	2.18		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD50	11637	886	834	985	1034	1090	1011	1000	992	962	986	937	920		
	CDD65	6162	421	414	520	584	625	561	535	527	512	521	487	455		
	CDH74	65042	3432	3900	5689	7416	7935	6511	5552	5482	5243	5276	4726	3880		
(13)	CDH80	24488	1001	1400	2345	3576	3795	2634	1840	1729	1680	1761	1587	1140		
(14)	Wind	WSAvg	2.4	2.2	2.2	2.3	2.4	2.7	2.5	2.8	2.6	2.3	2.2	2.4		
(15) Precipitation	PrecAvg	84.0	0.4	0.2	0.5	0.9	5.0	12.3	17.2	18.2	14.1	8.0	5.2	2.2		
	PrecMax	131.4	1.5	0.7	1.8	2.6	18.5	28.5	68.7	46.8	34.8	16.2	13.9	5.7		
	PrecMin	60.3	0.0	0.0	0.0	0.0	0.6	3.8	3.9	8.3	1.4	1.2	0.5	0.0		
	PrecStd	19.2	0.4	0.2	0.6	1.0	5.0	7.6	15.7	9.3	8.9	4.3	3.9	2.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.7	91.8	93.6	96.6	97.1	94.9	92.7	91.7	92.1	91.7	91.8	90.8		
		MCWB	76.1	76.1	76.8	78.2	78.8	79.8	80.1	80.2	79.8	78.0	77.7	76.6		
	2%	DB	87.8	90.0	92.3	95.0	95.4	93.0	90.9	90.0	90.3	90.1	90.3	89.0		
		MCWB	75.7	76.0	76.8	78.1	78.9	80.1	79.9	79.7	79.6	78.3	77.8	76.9		
	5%	DB	86.5	88.8	91.0	93.7	94.0	91.6	89.4	88.7	89.0	89.0	88.9	87.4		
		MCWB	75.3	75.8	76.6	78.1	79.1	80.0	79.7	79.5	79.4	78.4	77.7	76.8		
	10%	DB	85.2	87.3	89.5	92.4	92.4	90.0	87.9	87.2	87.4	87.7	87.5	85.8		
		MCWB	75.0	75.5	76.3	77.9	79.2	79.8	79.3	79.3	79.3	78.2	77.6	76.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.7	80.2	80.6	81.9	82.7	82.5	81.9	81.8	81.7	81.5	80.8	80.1		
		MCDB	85.5	87.4	89.5	91.9	91.6	91.0	89.6	88.5	88.6	88.0	87.9	86.3		
	2%	WB	77.6	78.4	79.1	80.7	81.6	81.5	81.0	80.9	80.8	80.5	79.8	79.0		
		MCDB	84.6	86.4	88.1	90.9	90.6	89.7	88.2	87.2	87.5	86.7	86.8	85.5		
	5%	WB	76.9	77.4	78.4	79.8	80.9	80.8	80.3	80.2	80.1	79.8	79.0	78.1		
		MCDB	84.0	85.5	87.3	89.9	89.7	88.7	87.1	86.1	86.4	85.9	85.9	84.4		
	10%	WB	76.0	76.5	77.4	79.1	80.2	80.2	79.7	79.7	79.6	79.1	78.3	77.3		
		MCDB	83.0	84.6	86.3	89.0	88.7	87.6	86.0	85.4	85.5	85.0	84.8	83.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.7	14.9	15.5	15.9	14.4	12.3	10.9	10.0	10.8	12.0	12.8	12.5		
		MCDBR	15.5	17.2	17.7	17.8	16.8	14.8	13.8	12.8	13.3	14.3	15.3	15.0		
	5% WB	MCWBR	6.7	7.2	7.0	6.5	5.6	5.1	5.2	4.9	4.8	5.2	5.6	6.0		
		MCWBR	13.8	14.9	15.6	16.0	14.7	13.1	12.3	11.1	12.1	12.2	13.2	12.8		
(40) Clear-Sky Solar Irradiance	taub	taub	0.420	0.418	0.450	0.475	0.475	0.467	0.462	0.471	0.466	0.476	0.439	0.439		
		taud	2.415	2.407	2.319	2.254	2.282	2.321	2.335	2.303	2.310	2.264	2.385	2.366		
	Ebn at Noon	Ebn at Noon	278	283	276	267	263	262	264	265	267	262	269	267		
		Edh at Noon	36	38	42	45	43	41	41	43	42	43	37	37		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1325	1568	1770	1993	1812	1630	1440	1387	1472	1407	1315	1182		
	RadStd	RadStd	80	137	123	108	160	161	180	140	144	116	119	102		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (16 neighbors)	+0.58	+0.85	N/A	+0.81	+0.72	+0.97	N/A	N/A	+284	+284

Nomenclature: See separate page